

YOUR LOWER RESOLUTION IS YOUR BEST WEAK MODEL: CROSS-RESOLUTION GUIDANCE FOR TINY PIXEL-ART DIFFUSION

Anonymous authors

Paper under double-blind review

ABSTRACT

Text-to-image models fail at the smallest images people draw on purpose: 12–24 px RGBA game sprites, where every pixel is a design decision and a large generator followed by downscaling or learned pixelization blurs exactly those decisions. We train a pixel-space diffusion model over a ladder of resolution buckets and show that it contains its own best weak model: the same network queried under a *lower* resolution label (optionally through an early snapshot) yields a structure-aligned, lower-contrast prediction whose difference from the target-resolution prediction is a precise guidance direction. The effect is directional—a higher or a random bucket hurts—and it is not a denoising-error correction: the MSE-optimal extrapolation weight never exceeds 1.07, on forward-noised and on self-rolled states alike. We then internalize this reference with a guidance-free-training objective, so that a single forward pass reproduces and surpasses the two-pass rule without storing any extra weights; with an empty-caption or a random-condition reference, the identical recipe is 39% and 10% worse. At 16 px the internalized model lowers FD from 11.28 (best-tuned CFG) to 7.23 at one network evaluation, and to 6.25 when combined with independent-condition guidance, 16% below the strongest training-free baseline; it leads at 20 and 24 px as well (−30%, −34%), and an 8 px rung extends the rule to 12 px. Generating with gpt-image-2 or nano-banana-2 and then downscaling, applying PixelOE, or applying learned pixelization gives a 3–7× higher FD at 16 px. We also report where the idea does not transfer: SDXL’s original-size conditioning is not a usable reference.

1 INTRODUCTION

Game sprites are among the smallest images anyone draws on purpose: an item or a character on a 12–24 px RGBA canvas, built from hard edges, a handful of flat colour regions and a transparent background. At these sizes there is no texture in which to hide a mistake. One wrong pixel changes the silhouette, and one blended colour reads as blur. The obvious recipe—prompt a large text-to-image model and shrink its output, or run a pixelization method on it—produces something recognisable but not a sprite: the result keeps the soft shading and the many colours of an illustration (Figure 2). Measured against held-out sprites, such pipelines are 3–7× further from the data than the model we train here (Section 4.5).

We therefore train a dedicated generator: a 72.5M-parameter pixel-space UNet that denoises RGBA sprites directly, conditioned on a frozen CLIP text encoder and on a *resolution bucket* label that tells it which of seven canvas sizes it is drawing on. Every sprite is also shown, box-downsampled, at every lower bucket, so the buckets share one population of objects and differ only in grid size. With a recaptioned corpus, this model already beats the large-model pipelines by a wide margin. What remains is the familiar gap between a trained diffusion model and its data, which guidance is meant to close. Classifier-free guidance (CFG; Ho & Salimans, 2022) closes little of it here: at its best weight it stops at FD 11.28 against a real-data floor of 3.45 at 16 px.

Our starting observation is that a bucketed model already contains a good weak model of itself. Queried on the same noisy input and the same caption but under the label of a *lower* bucket (12 when drawing at 16 px), the network returns a prediction on the same grid, with the same layout, but with systematically lower local contrast. Using it as the reference of autoguidance-style extrapolation (Karras et al., 2024), $e = e_{\text{ref}} + w(e_{\text{strong}} - e_{\text{ref}})$, is a training-free guidance rule at CFG’s cost, and composing it with an early EMA snapshot strengthens it further. This reference is *directional*: a higher or a uniformly random bucket makes samples worse than CFG.

Our main contribution is to *internalize* this reference. Following guidance-free training (GFT; Chen et al., 2025), we reparameterize the model’s prediction as $\beta s_{\theta}(x_t, c, b, \beta) + (1 - \beta) \text{sg}[r(x_t, c, b_{\text{low}})]$, where the stop-gradient reference r is an early snapshot of the model queried under the lower bucket, and train it on the ordinary denoising target. At the optimum, s_{θ} with $\beta = 1/w$ is the guided prediction, so sampling needs one network evaluation per step and no stored snapshot. The training recipe is GFT’s; the reference is not, and the reference is what matters: with GFT’s empty-caption reference the identical recipe reaches 10.05, with an independent-condition reference 7.96, and with our cross-resolution reference 7.23, reproduced by an independent re-training (7.20).

Contributions.

- A pixel-space, resolution-bucketed text-to-sprite diffusion model for 12–24 px RGBA, evaluated with a pipeline-matched FD protocol, and the observation that its own lower-resolution belief is a strong, training-free guidance reference (Section 3).
- A one-pass internalization of that reference. At 16 px it lowers FD from 11.28 (best CFG) to 7.23 at 1 NFE per step, and to 6.25 ± 0.09 when combined with independent-condition guidance (ICG; Sadat et al., 2025b) at 2 NFE, 16% below ICG alone, the strongest training-free baseline we evaluated (Section 4.2). It also leads at 20 and 24 px (Section 4.4).
- Controls that isolate the reference as the source of the gain. They include plain GFT, an internalized ICG reference, an online lower-bucket reference, fine-tuning alone and a second training run (Section 4.3). They are complemented by an analysis of what a good reference needs: a lower (not higher or random) resolution, output space rather than embedding space, and degradation along a condition the model actually uses (Section 5).
- An external comparison on a shared protocol against two commercial text-to-image models combined with downscaling, PixelOE (Yeh, 2024) and learned pixelization (Wu et al., 2022) (Section 4.5), and a negative result: SDXL’s size conditioning does not give a usable reference.

2 RELATED WORK

Guidance references. CFG (Ho & Salimans, 2022) extrapolates the conditional prediction away from an unconditional one. Many variants keep that reference and reshape the update: guidance intervals (Kynkäänniemi et al., 2024), APG (Sadat et al., 2025a), frequency decoupling (Sadat et al., 2025c), CADs (Sadat et al., 2024), CFG++ (Chung et al., 2025) and feedback or stage-wise schedules (Koulischer et al., 2025; Jin et al., 2026). A second line replaces the reference. Autoguidance (Karras et al., 2024) uses a smaller or less-trained version of the same model. Training-free weak predictions come from perturbed self-attention (PAG, SAG, SEG; Ahn et al., 2024; Hong et al., 2023; Hong, 2024), sliding windows (Adaloglou et al., 2025), stochastic sub-networks (Chen et al., 2026), a random condition or a perturbed time embedding (ICG, TSG; Sadat et al., 2025b), or a semantically degraded text condition (CDG; Han et al., 2026). Other weak predictions come from auxiliary or internal heads (Zhou et al., 2026; Yuan et al., 2026; Fu et al., 2026). CDG is closest in spirit: it too degrades one condition of the same network. It degrades the *text*, whereas we degrade the *resolution label* and keep the caption fixed. We run CDG, ICG, TSG, SEG and PAG under our protocol (Table 2).

Guidance without a second pass. Guidance distillation (Meng et al., 2023) regresses a student onto a guided teacher. Model guidance (Tang et al., 2025) and guidance-free training (Chen et al., 2025) instead change the training objective so that its fixed point is the guided prediction. GFT does this with a stop-gradient unconditional reference, a scalar β input, and the true noise as the target, and reports parity with CFG. We use GFT’s objective unchanged and swap its reference. Our β input reuses the guidance-scale embedding of latent consistency models (Luo et al., 2023).

Resolution conditioning. Multi-resolution training with a size or bucket label is standard: aspect-ratio bucketing (Anlatan (NovelAI), 2022), SDXL’s micro-conditioning (Podell et al., 2024), Matryoshka diffusion (Gu et al., 2024), FiT (Lu et al., 2024) and simple diffusion (Hooeboom et al., 2023). SDXL documents that a small `original_size` reproduces the blur of low-resolution training images, and common practice passes it to the negative branch of CFG (von Platen et al., 2022). We test this as a reference in Section 5 and find no benefit. To our knowledge, no prior work uses a model’s own lower-resolution label as a guidance reference or internalizes one.

Pixel art. Most pixel-art work converts an existing image, by optimisation (Gerstner et al., 2012), contrast-aware downscaling (PixelOE; Yeh, 2024) or learned pixelization (Han et al., 2018; Wu et al., 2022). Generative approaches include GANs for character sprites (Coutinho & Chaimowicz, 2022), pose-conditioned sprite sheets (Hsieh et al., 2024), a palette-conditioned small DDPM (Li, 2025), SD- π XL, which optimises one image by score distillation from SDXL (Binninger & Sorkine-Hornung, 2024), and community LoRAs such as Pixel Art XL (nerijs, 2023). We compare against conversion pipelines applied to strong generators; SD- π XL costs about 8 GPU-hours per sprite (Section 4.5). For evaluation we follow Stein et al. (2023) and use DINOv2 features (Oquab et al., 2024) rather than Inception (Heusel et al., 2017; Parmar et al., 2022).

3 METHOD

3.1 A RESOLUTION-BUCKETED PIXEL-SPACE SPRITE MODEL

We generate RGBA sprites on an $R \times R$ canvas, $R \in \{12, 16, 20, 24\}$, from a caption c . The denoiser $\epsilon_\theta(x_t, t, c, b)$ is a UNet (diffusers UNet2DConditionModel; 72.5M parameters; channel widths 128/256/512; two cross-attention stages at R and $R/2$; no attention at $R/4$ except the mid block). It operates directly on the 4-channel image. Text enters through cross-attention to the 77 token states of a frozen CLIP ViT-B/32 text encoder (Radford et al., 2021). The resolution bucket $b \in \mathcal{B} = \{12, 16, 20, 24, 32, 48, 64\}$ enters as a class embedding added to the timestep embedding, and it is never dropped. We train with ϵ -prediction (Ho et al., 2020), a cosine schedule with $T = 1000$, AdamW at a constant learning rate of $5 \cdot 10^{-5}$, EMA 0.999 and 10% caption dropout. We sample with 100 ancestral DDPM steps from the EMA weights.

Data. The corpus combines sprites from OpenGameArt (OpenGameArt.org, 2025) with the Kenney and Liberated Pixel Cup packs. Each sprite is centred on a transparent canvas at its native bucket (the smallest bucket that fits it) with a hard alpha threshold. It is additionally fed, premultiplied-box-downsampled, to every bucket in $\{12, 16, 20, 24\}$ at least 25% below its native size. The low buckets therefore see the same population of objects as the native ones. All 5,928 evaluation sprites are excluded (180,533 training rows including the low-bucket copies). The original BLIP captions (Li et al., 2022) were nearly uninformative: 15.4% of rows read “a pixel art sprite” and only 23.5% are distinct. We recaptioned all 43,342 unique images with gemini-3-flash (Google DeepMind, 2025), presented as image grids. This yields usable captions for 94.8% of them, 98.6% distinct, with a median of 12 words (Appendix B). The main model **v7r** is trained from scratch for 80k steps on the recaptioned corpus. **v7h** is the same recipe on the BLIP captions and is used only where stated.

3.2 THE LOWER RESOLUTION AS A WEAK REFERENCE

All guidance rules we compare combine the target-resolution prediction $e_{\text{strong}} = \epsilon_\theta(x_t, t, c, b_R)$ with a reference e_{ref} computed on the same x_t :

$$\tilde{e} = e_{\text{ref}} + w(e_{\text{strong}} - e_{\text{ref}}). \quad (1)$$

CFG uses $e_{\text{ref}} = \epsilon_\theta(x_t, t, \emptyset, b_R)$. Autoguidance (Karras et al., 2024) uses an early EMA snapshot θ_{snap} under the same condition. A bucketed model offers a third choice with no extra weights: the same network, input and caption under a *lower* bucket,

$$e_{\text{ref}}^{\text{label}} = \epsilon_\theta(x_t, t, c, b_{\text{low}}), \quad e_{\text{ref}}^{\text{comp}} = \epsilon_{\theta_{\text{snap}}}(x_t, t, c, b_{\text{low}}), \quad b_{\text{low}} < b_R. \quad (2)$$

We use $b_{\text{low}} = 12$ at 16 px and $b_{\text{low}} = 16$ at 20 and 24 px. The *composed* reference evaluates the snapshot under the lower label. It is a single weak forward pass, not a sum of two guidance terms.

Both rules cost 2 NFE per step, like CFG, and we use $w = 2$ (label) and $w = 1.5$ (composed). Because only the label differs, the reference is structure-aligned with the strong prediction by construction. On v7h, samples drawn from the 12 px belief alone at 16 px have about the strong model’s colour count (78 vs. 73 median colours) but much lower local contrast (mean adjacent $|\Delta\text{RGB}|$ 21.6 vs. 32.0; real sprites 30.2; Appendix E). For v7r, θ_{snap} is the 10k-step EMA of an independent 10k-step run with v7r’s recipe; v7r itself kept snapshots only every 20k steps.

3.3 INTERNALIZING THE REFERENCE

The training-free rule needs two passes per step and, in its composed form, a stored snapshot. We fold it into the weights with the objective of GFT (Chen et al., 2025), replacing GFT’s unconditional reference by the composed cross-resolution reference. The network gains a scalar input $\beta \in (0, 1]$, and its noise prediction during training is parameterized as

$$\hat{\epsilon}_{\beta}(x_t, t, c, b) = \beta s_{\theta}(x_t, t, c, b, \beta) + (1 - \beta) \text{sg}[r(x_t, t, c)], \quad r = \epsilon_{\theta_{\text{snap}}}(x_t, t, c, b_{\text{low}}(b)), \quad (3)$$

trained with $\mathcal{L} = \mathbb{E} \|\hat{\epsilon}_{\beta} - \epsilon\|^2$ on ordinary forward-noised data. Here sg is the stop-gradient and $b_{\text{low}}(b)$ is the reference rung (12 for 16, 16 for 20 and 24). Because the reference is fixed given (x_t, c) , the population optimum satisfies $\beta s^* + (1 - \beta)r = \mu := \mathbb{E}[\epsilon \mid x_t, c, b]$, i.e.

$$s^*(x_t, t, c, b, \beta) = r + \frac{1}{\beta}(\mu - r). \quad (4)$$

Setting $\beta = 1/w$ at inference therefore produces, in *one* forward pass, the guided rule of Eq. equation 1 with reference r (with the Bayes denoiser μ in place of the finite network). The snapshot is used only to compute training targets and is not needed at inference. Section 5 explains why we use this reparameterization rather than a learned corrector: a module trained with the plain denoising loss cannot learn to extrapolate away from this reference.

Implementation. β is embedded like LCM’s guidance scale (Luo et al., 2023): a 256-d sinusoidal embedding of 1000β , mapped by a *zero-initialized* linear layer (diffusers `time_cond_proj`, 32,768 parameters) and added to the timestep features. At step 0 the model is therefore exactly v7r. We draw $\beta \sim \mathcal{U}(0.4, 1)$, set $\beta = 1$ for a random 25% of rows (this keeps the plain model trained) and for all rows of the 12 px bucket (no lower rung). We fine-tune from v7r for 10k steps with v7r’s data, batch sizes, optimizer and EMA. The reference is a no-gradient forward of the frozen snapshot, so training costs one extra forward pass for the rows with $\beta < 1$. At inference we use $w = 1.5$ ($\beta = 2/3$) with 1 NFE per step. We call this model *internalized cross-resolution guidance*.

3.4 COMBINING WITH INDEPENDENT-CONDITION GUIDANCE

The internalized reference degrades the resolution axis. ICG (Sadat et al., 2025b) degrades the text axis: its reference replaces the caption embedding by i.i.d. Gaussian noise with the embedding’s standard deviation, redrawn at every step. The two compose at 2 NFE per step:

$$\tilde{\epsilon} = s_{\theta}(x_t, t, \hat{c}, b_R, \beta) + w_{\text{ICG}}(s_{\theta}(x_t, t, c, b_R, \beta) - s_{\theta}(x_t, t, \hat{c}, b_R, \beta)), \quad \hat{c} \sim \mathcal{N}(0, \text{std}(c)^2 I), \quad (5)$$

where both passes use the internalized model at the same β . We use internal weight $1/\beta = 1.25$ and $w_{\text{ICG}} = 1.5$. Stacking plain CFG on the internalized model instead hurts (8.07–8.34 vs. 7.63 on v7h, seed 0; Appendix E).

4 EXPERIMENTS

4.1 PROTOCOL

Matched FD. For every configuration we draw one sample for each of 3,000 held-out captions. These are the recaptions of 3,000 held-out sprites, so the prompt distribution matches the data. Samples are generated in three chunks of 1,000 with seeds $s, s+1, s+2$; “seed s ” in the tables refers to this. The reference set is 3,000 further real sprites, disjoint from the held-out ones and from training, passed through the same canvas preprocessing as the training data. Our primary metric is the Fréchet distance of DINOv2-small CLS features (Oquab et al., 2024; Stein et al., 2023) (FD; lower is better). Features are computed at the native resolution R : images are composited on white

Table 1: Matched FD-DINOv2 at 16 px on the recaptioned model v7r (3,000 held-out captions; floor 3.45). NFE per sampling step. “Ours” rows marked † use no training beyond v7r; the internalized rows fine-tune v7r for 10k steps. CLIP and R@1 on the seed-0 samples. Re-training: an independent second fine-tuning with the same recipe.

Method	w	NFE	seeds 0 / 1 / 2	FD mean $\pm$ sd	CLIP / R@1
Real held-out sprites (floor)	—	—	—	3.45	27.66 / 22.6%
CFG, training default	4	2	20.18 / — / —	—	27.96 / 24.7%
CFG, best weight	1.5	2	11.10 / 10.85 / 11.89	11.28 $\pm$ 0.54	27.68 / 23.1%
CDG (Han et al., 2026)	1.5	2	10.36 / — / —	—	27.43 / 21.7%
Autoguidance, 10k snapshot (Karras et al., 2024)	1.5	2	9.20 / 10.04 / 10.09	9.78 $\pm$ 0.50	—
ICG (Sadat et al., 2025b)	1.5	2	7.14 / 7.37 / 7.87	7.46 $\pm$ 0.37	27.72 / 23.4%
Lower-bucket label reference†	2	2	8.81 / 8.83 / 8.93	8.86 $\pm$ 0.06	27.05 / 18.7%
Composed reference† (snapshot + label)	1.5	2	8.22 / 8.59 / 8.70	8.50 $\pm$ 0.25	27.32 / 20.3%
GFT, empty-caption reference (Chen et al., 2025)	1.5	1	9.63 / 10.42 / 10.09	10.05 $\pm$ 0.40	[TODO: CLIP]
Internalized cross-resolution (ours)	1.5	1	6.79 / 7.68 / 7.22	7.23 $\pm$ 0.45	27.50 / 22.0%
re-training	1.5	1	7.36 / 6.87 / 7.36	7.20 $\pm$ 0.28	—
Internalized + ICG (ours)	1.25, 1.5	2	6.17 / 6.23 / 6.34	6.25 $\pm$ 0.09	[TODO: CLIP]
re-training	1.25, 1.5	2	6.24 / 6.67 / 6.03	6.31 $\pm$ 0.33	—

and nearest-neighbour upsampled to 224 px with ImageNet normalization. Scoring the 3,000 held-out real sprites against the reference gives the *floor*: 3.37, 3.45, 12.14 and 12.96 at 12, 16, 20 and 24 px. Seed-to-seed standard deviations are 0.06–0.54 at 16 px. Two independent trainings of the same recipe differ by about 0.6 at a single seed and by less than 0.1 in their three-seed means. Differences of that order are not interpreted. We report mean $\pm$ sample standard deviation over three seeds where marked, otherwise seed 0. *Best CFG* is the lowest-FD CFG weight at each resolution: $w = 1.5$ at 16 px (sweep 11.10 / 12.20 / 15.45 / 20.18 for $w = 1.5/2/3/4$), $w = 2$ at 12 and 20 px, and $w = 3$ at 24 px. We also report CLIP ViT-B/32 image–text similarity (100 cos) and retrieval R@1 of the own caption among 100.

4.2 MAIN RESULTS AT 16 PX

Table 1 gives the 16 px results. Recaptioning already makes the CFG baseline stronger than on BLIP captions (best CFG 11.28 vs. 12.49 on v7h). Among training-free references the cross-resolution ones clearly beat CFG (–21% label, –25% composed), autoguidance (–13%) and CDG (–7%, seed 0), but on this model they are beaten by ICG (–34%). The internalized model reaches 7.23 at *one* network evaluation per step: –36% against best CFG, 1.27 below the two-pass composed rule it was trained from (lower at every seed), and level with ICG (7.46) at half ICG’s cost. An independent re-training gives 7.20 ± 0.28 . Combined with ICG (Eq. 5), it reaches **6.25 $\pm$ 0.09**: 16% below ICG at equal cost and 45% below best CFG. The re-training gives 6.31 ± 0.33 , so all six values (two trainings $\times$ three seeds) lie below ICG’s best seed (7.14). The combination is sensitive to its weights, and 1.25/1.5 is the best of the seed-0 neighbourhood we evaluated (internal 1.15/1.25/1.35 with ICG 1.5: 6.39 / 6.17 / 6.45; internal 1.25 with ICG 1.25/1.75: 6.27 / 7.40). The internalized model alone is sensitive to w as well (1.35 / 1.5 / 1.75: 8.00 / 6.79 / 8.12, seed 0). Qualitatively (Figure 1), CFG samples show bleeding colours and noisy interiors, and the internalized rows are cleaner and closer to the real sprites’ flat regions and dark outlines. Text alignment of the internalized model (CLIP 27.50, R@1 22.0%) is close to real sprites (27.66, 22.6%) and above the training-free cross-resolution rules, whose same-caption reference removes the text term that CFG adds (R@1 18.7% for the label reference).

Table 2 lists all training-free references we evaluated. The two that perturb the self-attention computation (SEG, PAG) and the time-embedding perturbation (TSG) are worse than tuned CFG. At 16 px, the mid-block self-attention sees only 4×4 tokens, which leaves little to perturb. CDG improves slightly on CFG. ICG is the only strong external reference; its random-*bucket* variant is harmful (Section 5).

4.3 THE REFERENCE IS WHAT MAKES INTERNALIZATION WORK

GFT’s recipe alone does not explain the result. Table 3 changes only the training reference. With GFT’s empty-caption reference, the identical recipe internalizes CFG and reaches 10.05, 39% worse



Figure 1: 16px samples from v7r (seed 0; same held-out caption and initial noise in each column; nearest-neighbour upsampled on a checkerboard; FD from Table 1 in parentheses). Captions, left to right: ear of corn; magic wand with a red orb; goblin with a club; green bipedal creature with a pink tongue; magnifying glass with a green lens; magenta slime; potion bottle with purple liquid; cowboy with a star; large fire; baked pie (full captions in Appendix H). Columns were chosen from the first 60 held-out prompts for legibility of the real sprite, identically for all rows.

Table 2: Training-free guidance references on v7r at 16px (seed 0, 2 NFE per step, all used in Eq. equation 1 in place of CFG; best CFG 11.10 at this seed). Implementations follow the original papers; exact settings in Appendix C.

Method	Reference e_{ref}	Weights / settings $\rightarrow$ FD	Best
CFG	empty caption	w 1.5 / 2 / 3 / 4: 11.10 / 12.20 / 15.45 / 20.18	11.10
CDG (Han et al., 2026)	content tokens $\rightarrow$ empty, padding kept	w 1.5 / 2 / 3: 10.36 / 10.76 / 11.02	10.36
ICG (Sadat et al., 2025b)	Gaussian condition, redrawn per step	w 1.25 / 1.5 / 2: 7.68 / 7.14 / 10.04	7.14
ICG-label	uniformly random bucket	w 1.5 / 2: 12.23 / 12.68	12.23
TSG (Sadat et al., 2025b)	noised time embedding	SD: w 1.5 / 2.5 / 4: 17.73 / 27.02 / 37.28; DiT: 17.73	17.73
SEG (Hong, 2024)	blurred self-attention queries	mid $\sigma=\infty$: 12.80; d1+u1 σ 1 / 2: 16.88 / 26.68	12.80
PAG (Ahn et al., 2024)	identity self-attention (mid)	w 1.5 / 2: 12.38 / 12.21	12.21
Autoguidance	10k EMA snapshot	w 1.5: 9.20	9.20
Ours: label	same network, bucket 12	w 2: 8.81	8.81
Ours: composd	snapshot under bucket 12	w 1.5: 8.22	8.22
ICG + lower bucket	Gaussian condition and bucket 12	w 1.25 / 1.5 / 2: 7.22 / 7.16 / 11.68	7.16

than ours and only 11% better than best CFG. Internalizing ICG’s Gaussian-condition reference gives 7.96, 10% worse than ours and worse than ICG applied at sampling time (7.46). Of the references we tried, only the cross-resolution one yields a 1-NFE model that beats its own two-pass rule. Within the cross-resolution family, the early snapshot matters: an online reference (the network itself at $\beta = 1$ under the lower bucket) reaches only 9.13 against 7.36 for the snapshot reference in the same re-training run. Fine-tuning by itself contributes little: with internal guidance switched off ($\beta = 1$), ICG on the GFT model moves from 7.14 to 7.02. With ICG on top, our reference is 0.46 better than the empty-caption reference at the same internal weight (6.17 vs. 6.63). Longer fine-tuning does not help. Our reference reaches 7.47 after 20k steps (three seeds) against 7.23 at 10k, and the re-training’s curve is 7.66 / 7.39 / 7.36 at 5k / 7.5k / 10k (seed 0). We therefore use

Table 3: What the internalized model’s gain comes from (16 px, v7r). All fine-tuned rows use the identical recipe (init v7r, 10k steps, $\beta \sim \mathcal{U}(0.4, 1)$ with 25% $\beta = 1$) and differ only in the training reference r of Eq. equation 3. Top: 1 NFE at $\beta = 1/1.5$. Bottom: the same models with ICG 1.5 on top (2 NFE). *compared with the re-training run of the same recipe (7.36 at 10k, seed 0).

Training reference r	Sampling	seeds	FD
Snapshot under lower bucket (ours)	1 NFE, $w = 1.5$	3	7.23 ± 0.45 (re-training 7.20 ± 0.28)
Online network under lower bucket, no snapshot	1 NFE, $w = 1.5$	1	9.13*
Online network, Gaussian condition (internalized ICG)	1 NFE, $w = 1.5$	3	7.96 ± 0.22
Online network, empty caption (= GFT)	1 NFE, $w = 1.5$	3	10.05 ± 0.40
none (v7r, no fine-tuning)	ICG 1.5	3	7.46 ± 0.37 (seed 0: 7.14)
Empty caption (GFT), internal guidance off ($\beta = 1$)	ICG 1.5	1	7.02
Empty caption (GFT), internal $w = 1.25$	ICG 1.5	1	6.63
Snapshot under lower bucket (ours), internal guidance off	ICG 1.5	1	6.63
Snapshot under lower bucket (ours), internal $w = 1.25$	ICG 1.5	3	6.25 ± 0.09 (seed 0: 6.17)

Table 4: Matched FD at 20 and 24 px (v7r; floors 12.14 / 12.96). “3s”/“2s”: mean over three/two seeds; otherwise seed 0. Reference rung 16 at both resolutions.

Method	NFE	20 px	24 px
Best CFG ($w = 2 / w = 3$)	2	34.21 (3s)	65.87 (3s)
ICG ($w = 1.5 / 2$)	2	28.30 / 29.73	48.82 / 46.21
Autoguidance, 10k snapshot	2	26.54 (3s)	52.71
Lower-bucket label reference, $w = 2$ (ours, training-free)	2	27.73 (3s)	49.99
Composed reference, $w = 1.5$ (ours, training-free)	2	25.64 (3s)	50.26
Internalized, $w = 1.5$ (ours)	1	24.06 ± 0.35 (3s)	43.35 (2s) [TODO: 3rd seed]
Internalized $w = 1.25 + \text{ICG } 1.5$ (ours)	2	27.39	42.77 ± 1.1 (3s)

10k steps throughout; this choice was made on the evaluation metric (Section 6). The internalization also holds on the BLIP-caption model v7h (7.23 ± 0.34 at 1 NFE vs. composed 7.53 ± 0.19 at 2 NFE; Appendix E).

4.4 OTHER RESOLUTIONS: 20, 24 AND 12 PX

At 20 and 24 px (Table 4) the internalized model at 1 NFE is the best single method: -30% and -34% against best CFG, ahead of ICG at either weight and of every training-free cross-resolution rule. Unlike at 16 px, ICG does not add to it. At 20 px the combination is worse than the internalized model alone (27.39 vs. 24.06; 25.6–25.8 with ICG 1.25). At 24 px it is within one seed standard deviation (42.77 ± 1.1 vs. 43.35). The per-resolution recommendation is therefore the combination at 16 px and the internalized model alone at 20 and 24 px. We attribute the difference to ICG, whose own advantage over the cross-resolution rules is specific to 16 px (at 20 px it is behind all of them), but we have not isolated the cause.

12 px via an 8 px rung. v7r has no bucket below 12 px, so neither the training-free rule nor the internalization applies at 12 px. There, best CFG gives 8.82 (three seeds; floor 3.37) and autoguidance 8.36. We add an 8 px rung (**v7r8**): the class embedding grows to eight rows, the new row is initialized from the 12 px row, every sprite of at least 10 px contributes an 8 px copy (44k rows), and v7r is fine-tuned for 20k steps. At 12 px the label reference with $b_{\text{low}} = 8$ reaches 7.50 at $w = 2.5$ (9.77 / 8.15 at $w = 1.5/2$), against 8.40 for the best CFG weight on the same model (8.40 / 8.42 / 8.53 at $w = 1.5/2/2.5$), i.e. -11% at seed 0 and still improving with w (**[TODO: $w = 3/3.5$ and seeds 1–2]**). Adding the rung does not hurt 16 px (CFG 10.58 vs. 11.10 on v7r, seed 0). The internalized model on v7r8 (12 px reference rung 8) is training: **[TODO: 12 px internalized v7r8, three seeds, and its 16 px regression check]**.

4.5 EXTERNAL SYSTEMS

Table 5 compares against the most common practical alternative: render with a strong text-to-image model, then convert to a sprite. Every system receives the same recaption plus a fixed suffix (“pixel

Table 5: External comparison on the $n = 200$ protocol: the first 200 held-out recaptions, FD against the same 3,000-sprite reference set. At $n = 200$ FD is biased upward (floor 26.53 at 16 px vs. 3.45 at $n = 3000$), so only the comparison within a column is meaningful. Large-model renders are produced once at 1024 px and then converted to each resolution.

Method	16 px	20 px	24 px
Real held-out sprites ($n = 200$ floor)	26.53	94.67	98.80
Ours: internalized + ICG (16 px configuration)	27.81	—	—
Ours: internalized, 1 NFE	29.60	107.03	129.15
Our model v7r, best CFG	30.00	112.60	161.21
Our model v7r, ICG	31.72	116.72	143.85
gpt-image-2 (OpenAI, 2026) + downscale	82.23	317.62	428.51
gpt-image-2 + PixelOE (Yeh, 2024)	146.32	282.51	357.97
gpt-image-2 + Make Your Own Sprites (Wu et al., 2022)	155.20	336.66	431.21
nano-banana-2 (Google DeepMind, 2026) + downscale	87.35	330.47	409.57
nano-banana-2 + PixelOE	150.79	301.03	366.36
nano-banana-2 + Make Your Own Sprites	186.95	386.73	441.40
SDXL + Pixel Art XL LoRA (Podell et al., 2024; nerijs, 2023) + downscale	[TODO:]	[TODO:]	[TODO:]
SDXL + Pixel Art XL LoRA + PixelOE	[TODO:]	[TODO:]	[TODO:]



Figure 2: “Generate large, then shrink” vs. ours at 16 px (same captions as Figure 1; $n = 200$ FD in parentheses). The 1024 px renders (shown as thumbnails) often follow the caption closely, but after conversion they keep soft shading, anti-aliased edges and many colours, which are absent from the data.

art sprite of a single object, plain flat white background, centred, no text”). The 1024 px output is cut out from its background by a border flood fill and converted to $R \times R$ in one of three ways. *Downscale* uses our own training preprocessing (premultiplied box filter, centred canvas, hard alpha). *PixelOE* (Yeh, 2024) uses contrast mode with the alpha taken by majority vote per cell. *Make Your Own Sprites* (Wu et al., 2022) uses the official weights with 4 px cells at a $4R$ input, reduced by 4. Our own rows are recomputed on the same 200 prompts. At 16 px our best configuration is 1.3 above the $n = 200$ floor, while the external pipelines are 55.7–160 above it: an FD 3.0–6.7 times ours. The ordering holds at 20 and 24 px. Learned pixelization is the weakest converter at these extreme ratios, and PixelOE is worse than plain downscaling at 16 px but better at 20–24 px. Figure 2 shows why FD separates the systems so clearly. The large models often follow the caption more literally (the corn, the frog), but after conversion their sprites keep soft shading, anti-aliased edges and

many colours, unlike the flat, outlined style of the data. We did not measure text alignment for the external systems ([**TODO: CLIP for external rows**]). SD- π XL (Binniger & Sorkine-Hornung, 2024) optimises each sprite by score distillation from SDXL. It took 8.2 GPU-hours per 16 px sprite in our measurement (2.95 s per step $\times$ 10,001 steps), against about 0.04 s per sprite for our 1-NFE sampler (42.5 s per 1,000 samples; Appendix F). An FD over 200 prompts is therefore out of reach, and we show only a small qualitative set ([**TODO: SD- π XL sprites on the recaptioned prompts; one of about 30 finished**]).

5 WHY A LOWER RESOLUTION IS A GOOD REFERENCE

This section asks what the reference must satisfy. Its results constrain the design of Section 3; they do not establish a mechanism.

Direction matters. If any wrong bucket acted as a generic unconditional-like reference, a higher or a random bucket would work as well as a lower one. Neither does. On v7r at 16 px, a bucket drawn uniformly at random at every step (ICG applied to the resolution label) gives 12.23 / 12.68 at $w = 1.5/2$. That is *worse* than best CFG (11.10, seed 0), whereas the lower bucket gives 8.81. On v7h, every higher-bucket reference is worse than best CFG at every resolution (16 px: buckets 20 / 24 / 64 give 17.14 / 17.27 / 19.34 vs. 12.24; Appendix E). On v7h the label also acts as a monotone contrast knob: the pure belief’s local contrast rises from 21.6 to 31.9 to 40.9 for buckets 12, 24 and 64 at 16 px. Only its low-contrast end is useful. Isotropic perturbation of the same embedding (TSG) is harmful at every setting we tried (17.73–37.28; Table 2).

The guidance is not a denoising-error correction. A tempting alternative to Eq. equation 3 is a learned module that reads the lower-bucket belief and corrects the strong prediction, trained with the plain denoising loss. Any such module converges to μ and can extrapolate away from r only as far as the finite network’s residual is linearly visible in $d = e_{\text{strong}} - r$. The best such extrapolation weight is directly measurable: $w^*(t) = \mathbb{E}[d \cdot (\epsilon - r)] / \mathbb{E}[d \cdot d]$. This is unbiased because $\epsilon - \mu$ has zero mean given x_t . On v7h, on 3,000 held-out sprites at 10 timesteps (and 3,000 training sprites at 16 px), w^* lies in 0.83–1.07 for the label, snapshot and composed references at 16, 20 and 24 px. It is often below 1 on held-out data, and extrapolating to $w = 1.5$ increases the denoising error everywhere (by 1–95%). On the model’s own trajectories, obtained by noising a real sprite and running the deployed sampler for $k \in \{1, 5, 20\}$ steps, w^* in x_0 -space stays at or below 1.07 for $k \geq 5$ at all three resolutions (0.64–0.78 is typical at $k = 5$; Appendix D). Guidance with $w \approx 1.5$ –2 improves FD while *increasing* per-sample regression error. It shapes the distribution, as CFG does, rather than correcting individual predictions. Hence a reparameterization whose fixed point is the guided rule (Eq. 4) is needed; a learned corrector is not enough.

The reference must act in output space. The cheapest form of 1-NFE extrapolation acts on the label embedding itself: $E[16] + (w - 1)(E[16] - E[12])$, used with a single conditional pass. It fails badly on v7h at 16 px: 39.03 at $w = 1.5$ and 414.27 at $w = 2$, against 16.39 without any guidance. The lower label is useful only as a second prediction that the update extrapolates away from, and the internalization of Section 3.3 keeps it in that role.

A reference must degrade along an axis the model uses. Recaptioning reverses the ranking of the text-axis and resolution-axis references. With BLIP captions (v7h), which the model largely ignores, ICG is weak (best 8.88 at $w = 1.25$; 11.29 at $w = 1.5$), and the cross-resolution rules are strong (label 8.52, composed 7.53, best CFG 12.49). With informative captions (v7r), ICG becomes the strongest training-free reference (7.46), and the cross-resolution rules still help but less (8.86 / 8.50 vs. best CFG 11.28). A random text condition is a weak reference only when the prediction depends on the text. The resolution label matters regardless of caption quality, since every sprite is trained at several buckets. The two axes are complementary at 16 px (Section 4.2), but only in the internalized form. Placing ICG’s condition and the lower bucket in the same training-free weak pass gives 7.16, no better than ICG alone (7.14, seed 0), and adding the snapshot to that pass is harmful (11.65).

Where it does not transfer: SDXL’s size conditioning. SDXL’s `original_size` micro-conditioning is the closest public analogue of a resolution label. We test it on 1,000 COCO-30k

captions at 1024 px (Euler, 30 steps; FD with DINOv2 ViT-L/14 against 2,000 real images, floor 507.05). CFG gives 662.22 / 644.18 / 648.44 at $w = 3/5/7$. Adding $1 \times (e_c - e_{c, \text{orig}=512})$ on top of CFG 5 gives 644.13 (no change), and with $\text{orig} = 256$ it gives 658.98 (worse). The size-conditioned reference without CFG collapses (969.68). The common `negative_original_size` practice gives 650.98. The only agreement with our model is directional: a weak branch that keeps the caption beats one that drops it. We believe that SDXL’s size condition encodes the blur of upsampled low-resolution training images rather than a coarser target grid, so it does not produce a structure-aligned, lower-contrast belief of the same image. The reference we propose is a property of models trained natively over a ladder of grids, not of size conditioning in general.

6 LIMITATIONS AND CONCLUSION

Limitations. (i) *Scope.* Everything is measured on models we trained, from one architecture family, on one corpus of game sprites. We have not tested a second domain, and SDXL’s size conditioning does not transfer. (ii) *Metric.* FD with DINOv2 on images of 12–24 px, upsampled to 224 px, assigns one sprite pixel to one or a few patches. It is sensitive to exactly the colour and contrast statistics that guidance changes. We report CLIP alignment for most 16 px rows, but we have no human study. The CLIP evaluator also shares its text tower with our model’s frozen encoder. (iii) *Tuning.* Guidance weights, the ICG combination, the reference rung and the 10k-step fine-tuning length were all selected on the same held-out FD that we report; there is no separate validation split. Seed and re-training variation (Section 4.1) is of the same order as some of the smaller differences, which we do not interpret. (iv) *Coverage.* 12 px needs an extra 8 px rung, and its internalized result is pending. Several 20/24 px rows are single-seed. The combination with ICG helps at 16 px but not at 20 px. (v) *Novelty of the objective.* The training objective is GFT’s; our claim is the reference and the evidence that it, not the objective, carries the gain.

Conclusion. A diffusion model trained over a ladder of resolutions contains its own best weak model: the same network under a lower resolution label. Extrapolating away from it is directional, acts in output space and is not an error correction. Folding it into the weights with a guidance-free-training objective gives a one-pass sampler that matches or beats every training-free guidance rule we tested at 16 px and is the best single method at 20 and 24 px. Combined with ICG it reaches 6.25 at 16 px, against 11.28 for best CFG and 82–187 for large-model-plus-conversion pipelines on the shared protocol.

REPRODUCIBILITY STATEMENT

Section 3.1 and Appendix A specify the architecture, data processing, training recipe and sampler. Equations equation 3–equation 5 and Section 3.3 give the full internalization objective with all hyperparameters. Section 4.1 and Appendix A specify the evaluation protocol: prompt set, chunked seeding, reference set, preprocessing and floors. Appendix C gives the exact settings of every guidance baseline, and Section 4.5 the conversion pipelines for the external systems. We will release code for training, sampling and evaluation, the recaptions, and the held-out split **[TODO: anonymous code link in the supplementary material]**. The raw sprite corpus carries per-asset licences and will be released as a list of sources after a licence audit (see the ethics statement).

ETHICS STATEMENT

The model generates small game sprites. It adds no capability beyond existing text-to-image systems, and its outputs (12–24 px) have little potential for targeted misuse. Its intended application raises the usual concerns of generative art tools, namely the displacement of pixel artists and the reproduction of styles present in the training corpus. The training data are community sprites from OpenGameArt and the Kenney and Liberated Pixel Cup packs. OpenGameArt submissions carry per-asset licences (CC0, CC-BY, CC-BY-SA, GPL, OGA-BY) that were not recorded at collection time, so a per-asset licence audit precedes any release of the corpus. Captions were produced by a commercial vision-language model; external baselines were generated through commercial APIs under their terms of use. No human subjects were involved.

AI USE STATEMENT

Generative AI tools were used in this work in the following ways. (a) As part of the method’s data pipeline, a vision-language model (gemini-3-flash) produced the training and evaluation captions (Section 3.1), and two commercial image generators served as external baselines (Section 4.5). (b) An LLM-based coding assistant was used to implement and run experiments, to search and verify the literature, and to draft and edit the manuscript. **[TODO: authors: complete the required disclosure per the ICLR 2027 AI policy, e.g. which ideas, analyses and text were AI-assisted and how they were verified]**. The authors reviewed all AI-assisted work and take responsibility for the final content, including all text, numbers and claims.

REFERENCES

- Nikolas Adaloglou, Tim Kaiser, Damir Iagudin, and Markus Kollmann. Guiding a diffusion model using sliding windows. In *British Machine Vision Conference (BMVC)*, 2025.
- Donghoon Ahn, Hyungwon Cho, Jaewon Min, Wooseok Jang, Jungwoo Kim, SeonHwa Kim, Hyun Hee Park, Kyong Hwan Jin, and Seungryong Kim. Self-rectifying diffusion sampling with perturbed-attention guidance. In *European Conference on Computer Vision (ECCV)*, 2024.
- Anlatan (NovelAI). NovelAI improvements on Stable Diffusion. NovelAI blog, <https://blog.novelai.net/novelai-improvements-on-stable-diffusion-e10d38db82ac>, October 2022. Origin of aspect-ratio bucketing; reference implementation at <https://github.com/NovelAI/novelai-aspect-ratio-bucketing>.
- Alexandre Binninger and Olga Sorkine-Hornung. SD- π XL: Generating low-resolution quantized imagery via score distillation. In *SIGGRAPH Asia 2024 Conference Papers*, 2024. doi: 10.1145/3680528.3687570.
- Chubin Chen, Jiashu Zhu, Xiaokun Feng, Nisha Huang, Chen Zhu, Meiqi Wu, Fangyuan Mao, Jiahong Wu, Xiangxiang Chu, and Xiu Li. S²-Guidance: Stochastic self-guidance for training-free enhancement of diffusion models. In *International Conference on Learning Representations (ICLR)*, 2026.
- Huayu Chen, Kai Jiang, Kaiwen Zheng, Jianfei Chen, Hang Su, and Jun Zhu. Visual generation without guidance. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 8275–8295, 2025.
- Hyunjin Chung, Jeongsol Kim, Geon Yeong Park, Hyelin Nam, and Jong Chul Ye. CFG++: Manifold-constrained classifier free guidance for diffusion models. In *International Conference on Learning Representations (ICLR)*, 2025.
- Flavio Coutinho and Luiz Chaimowicz. On the challenges of generating pixel art character sprites using GANs. In *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, volume 18, pp. 87–94, 2022. doi: 10.1609/aiide.v18i1.21951.
- Zixuan Fu, Chong Wang, Lanqing Guo, Kailai Zhou, Jiahao Nie, and Bihan Wen. A frozen pixel-space diffusion model can guide itself with its own samples. *arXiv preprint arXiv:2607.29122*, 2026.
- Timothy Gerstner, Doug DeCarlo, Marc Alexa, Adam Finkelstein, Yotam Gingold, and Andrew Nealen. Pixelated image abstraction. In *Proceedings of the 10th International Symposium on Non-Photorealistic Animation and Rendering (NPAR)*, pp. 29–36. Eurographics Association, 2012. Extended version: *Computers & Graphics* 37(5):333–347, 2013, doi:10.1016/j.cag.2012.12.007.
- Google DeepMind. Gemini 3 pro model card. <https://deepmind.google/models/model-cards/gemini-3-pro/>, November 2025. Documents the Gemini 3 model family; the gemini-3-flash API model was used for recaptioning.
- Google DeepMind. Gemini 3.1 flash image (nano banana 2) model card. <https://deepmind.google/models/model-cards/gemini-3-1-flash-image/>, February 2026.

- Jiatao Gu, Shuangfei Zhai, Yizhe Zhang, Josh Susskind, and Navdeep Jaitly. Matryoshka diffusion models. In *International Conference on Learning Representations (ICLR)*, 2024.
- Chu Han, Qiang Wen, Shengfeng He, Qianshu Zhu, Yinjie Tan, Guoqiang Han, and Tien-Tsin Wong. Deep unsupervised pixelization. *ACM Transactions on Graphics*, 37(6):243:1–243:11, 2018. doi: 10.1145/3272127.3275082. SIGGRAPH Asia 2018.
- Shilong Han, Yuming Zhang, and Hongxia Wang. Guiding diffusion models with semantically degraded conditions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 43653–43663, 2026.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022. A short version appeared at the NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Susung Hong. Smoothed energy guidance: Guiding diffusion models with reduced energy curvature of attention. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pp. 66743–66772, 2024.
- Susung Hong, Gyuseong Lee, Wooseok Jang, and Seungryong Kim. Improving sample quality of diffusion models using self-attention guidance. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 7462–7471, 2023.
- Emiel Hooeboom, Jonathan Heek, and Tim Salimans. simple diffusion: End-to-end diffusion for high resolution images. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, 2023.
- Cheng-An Hsieh, Jing Zhang, and Ava Yan. Sprite sheet diffusion: Generate game character for animation. *arXiv preprint arXiv:2412.03685*, 2024.
- Cheng Jin, Qitan Shi, and Yuantao Gu. Stage-wise dynamics of classifier-free guidance in diffusion models. In *International Conference on Learning Representations (ICLR)*, 2026.
- Tero Karras, Miika Aittala, Tuomas Kynkäänniemi, Jaakko Lehtinen, Timo Aila, and Samuli Laine. Guiding a diffusion model with a bad version of itself. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Felix Koulischer, Florian Handke, Johannes Deleu, Thomas Demeester, and Luca Ambrogioni. Feedback guidance of diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Tuomas Kynkäänniemi, Miika Aittala, Tero Karras, Samuli Laine, Timo Aila, and Jaakko Lehtinen. Applying guidance in a limited interval improves sample and distribution quality in diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *Proceedings of Machine Learning Research*, pp. 12888–12900, 2022.
- Lang Li. PixDiff-PIG: Palette-informed diffusion for pixel art generation. *Applied and Computational Engineering*, 211(1):217–224, 2025. doi: 10.54254/2755-2721/2026.tj30678.
- Zeyu Lu, Zidong Wang, Di Huang, Chengyue Wu, Xihui Liu, Wanli Ouyang, and Lei Bai. FiT: Flexible vision transformer for diffusion model. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 33160–33176, 2024.

- Simian Luo, Yiqin Tan, Longbo Huang, Jian Li, and Hang Zhao. Latent consistency models: Synthesizing high-resolution images with few-step inference. *arXiv preprint arXiv:2310.04378*, 2023.
- Chenlin Meng, Robin Rombach, Ruiqi Gao, Diederik P. Kingma, Stefano Ermon, Jonathan Ho, and Tim Salimans. On distillation of guided diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14297–14306, 2023.
- nerijs. Pixel art XL. <https://huggingface.co/nerijs/pixel-art-xl>, 2023. LoRA for Stable Diffusion XL 1.0 (base), Hugging Face model repository.
- OpenAI. GPT Image 2 (gpt-image-2). <https://developers.openai.com/api/docs/models/gpt-image-2>, 2026. API model page.
- OpenGameArt.org. OpenGameArt.org. <https://opengameart.org>, 2025. Community repository of freely licensed game art. Individual submissions carry per-asset licences (CC0, CC-BY, CC-BY-SA, GPL, OGA-BY); state the subset used and its licence terms.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research (TMLR)*, 2024.
- Gaurav Parmar, Richard Zhang, and Jun-Yan Zhu. On aliased resizing and surprising subtleties in GAN evaluation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11410–11420, 2022.
- Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. SDXL: Improving latent diffusion models for high-resolution image synthesis. In *International Conference on Learning Representations (ICLR)*, 2024.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pp. 8748–8763, 2021.
- Seyedmorteza Sadat, Jakob Buhmann, Derek Bradley, Otmar Hilliges, and Romann M. Weber. CADs: Unleashing the diversity of diffusion models through condition-annealed sampling. In *International Conference on Learning Representations (ICLR)*, 2024.
- Seyedmorteza Sadat, Otmar Hilliges, and Romann M. Weber. Eliminating oversaturation and artifacts of high guidance scales in diffusion models. In *International Conference on Learning Representations (ICLR)*, 2025a.
- Seyedmorteza Sadat, Manuel Kansy, Otmar Hilliges, and Romann M. Weber. No training, no problem: Rethinking classifier-free guidance for diffusion models. In *International Conference on Learning Representations (ICLR)*, 2025b.
- Seyedmorteza Sadat, Tobias Vontobel, Farnood Salehi, and Romann M. Weber. Guidance in the frequency domain enables high-fidelity sampling at low cfg scales. *arXiv preprint arXiv:2506.19713*, 2025c.
- George Stein, Jesse C. Cresswell, Rasa Hosseinzadeh, Yi Sui, Brendan Leigh Ross, Valentin Vilić, Zhaoyan Liu, Anthony L. Caterini, J. Eric T. Taylor, and Gabriel Loaiza-Ganem. Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Zhicong Tang, Jianmin Bao, Dong Chen, and Baining Guo. Diffusion models without classifier-free guidance. *arXiv preprint arXiv:2502.12154*, 2025.

Patrick von Platen, Suraj Patil, Anton Lozhkov, Pedro Cuenca, Nathan Lambert, Kashif Rasul, Mishig Davaadorj, Dhruv Nair, Sayak Paul, William Berman, Yiyi Xu, Steven Liu, and Thomas Wolf. Diffusers: State-of-the-art diffusion models. <https://github.com/huggingface/diffusers>, 2022. Micro-conditioning documentation: https://huggingface.co/docs/diffusers/api/pipelines/stable_diffusion/stable_diffusion_xl.

Zongwei Wu, Liangyu Chai, Nanxuan Zhao, Bailin Deng, Yongtuo Liu, Qiang Wen, Junle Wang, and Shengfeng He. Make your own sprites: Aliasing-aware and cell-controllable pixelization. *ACM Transactions on Graphics*, 41(6):193:1–193:16, 2022. doi: 10.1145/3550454.3555482. SIGGRAPH Asia 2022.

Shin-Ying Yeh. Detail-oriented pixelization based on contrast-aware outline expansion. <https://github.com/KohakuBlueleaf/PixelOE>, March 2024. Software; a classical (non-learned) image-processing pipeline.

Liangyu Yuan, Yufei Huang, Mingkun Lei, Tong Zhao, Ruoyu Wang, Changxi Chi, Yiwei Wang, and Chi Zhang. Improving diffusion generalization with weak-to-strong segmented guidance. *arXiv preprint arXiv:2603.20584*, 2026.

Xingyu Zhou, Qifan Li, Xiaobin Hu, Hai Chen, and Shuhang Gu. Guiding a diffusion transformer with the internal dynamics of itself. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

A IMPLEMENTATION AND PROTOCOL DETAILS

Table 6: Training and sampling configuration. v7h and v7r differ only in the caption files.

Denoiser	diffusers UNet2DConditionModel, 4 input/output channels (RGBA), widths 128/256/512, 2 layers per block; down: CrossAttn, CrossAttn, plain; up: plain, CrossAttn, CrossAttn; 72.5M parameters
Text	frozen CLIP ViT-B/32 text encoder, 77 tokens, last hidden states (512-d) as cross-attention context
Resolution	class embedding over $B = \{12, 16, 20, 24, 32, 48, 64\}$ added to the timestep embedding; never dropped
Diffusion	ϵ -prediction, $T = 1000$, cosine (squaredcos_cap_v2) schedule
Base training	from scratch, 80k steps, AdamW, constant lr $5 \cdot 10^{-5}$, EMA 0.999, caption dropout 10%
Batches	one bucket per batch, bucket drawn proportionally to its row count; batch 320 / 256 / 224 / 192 / 128 / 64 / 40 for 12 / 16 / 20 / 24 / 32 / 48 / 64 px
Snapshot	10k-step EMA of an independent 10k-step run with the same recipe (v7r); step-10k EMA of the run itself (v7h)
Internalization	init v7r, 10k steps, same optimizer/lr/EMA/batches; $\beta \sim \mathcal{U}(0.4, 1)$, $\beta=1$ w.p. 0.25 and for 12 px rows; reference rung 12 (for 16 px rows) and 16 (for 20 and 24 px rows); β via 256-d sinusoidal embedding of 1000β and a zero-initialized linear map into the timestep features (32,768 parameters)
Sampling	100-step ancestral DDPM, EMA weights, one sample per prompt

Canvas preprocessing. A sprite larger than the target side R is reduced with a premultiplied-alpha box filter, preserving aspect ratio; a sprite at most $R/2$ is enlarged by an integer nearest-neighbour factor. It is then centred on a transparent $R \times R$ canvas, and alpha is thresholded at 128 (pixels below become fully transparent, with RGB cleared). Generated samples are stored with the same hard alpha. The same function produces the FD reference set, the held-out floor set and the “downscale” external baselines.

FD. Real sprites are split once with a fixed shuffle into a 3,000-sprite reference set and a disjoint held-out pool. The first 3,000 held-out sprites supply the floor set and, through their recaptions, the 3,000 evaluation prompts. Each image is composited on white and resized to $R \times R$ (box filter; generated images are already $R \times R$). It is then nearest-neighbour upsampled to 224×224 , normalized with ImageNet statistics, and embedded by the CLS token of facebook/dinov2-small. FD is the Fréchet distance between Gaussian fits of the two feature sets (3,000 vs. 3,000; 200 vs. 3,000 for the external protocol). Samples are generated in chunks of 1,000 prompts, and chunk i uses seed $s + i$.

CLIP. The same saved samples are composited on mid-grey, nearest-neighbour upsampled to 224 px and scored with CLIP ViT-B/32 against the caption they were sampled from. We report 100 cos and R@1, the fraction of samples whose own caption ranks first among itself and 99 random held-out captions (chance 1%).

B RECAPTIONING

Table 7: Caption statistics before and after recaptioning (43,342 unique training and evaluation images).

	BLIP (v7h)	gemini-3-flash (v7r)
Distinct captions	23.5%	98.6%
Words per caption (median / mean)	8 / 7.6	12 / 12.1
Occurrences of the most frequent caption	7,669 (“a pixel art sprite”)	6 (“human”)
Share of data covered by the ten most frequent captions	$\approx 25\%$	$< 0.1\%$

Images were sent as numbered grids on a checkerboard (20 per request, 12 for retries). For each cell, the instruction asked for a caption of 6–15 words that names the object first and then its distinguishing colours, materials and parts, describes only what is visible, and avoids uninformative words (“pixel art”, “sprite”, “8-bit”, “icon”). Of the 43,342 unique images, 41,560 received a caption and 41,103 (94.8%) were usable. The remainder were declared an “unclear abstract shape” by the captioner and keep their BLIP caption. Across the three caption files (49,652 rows), 95% of rows were replaced. The total cost was about 18.7M tokens. The held-out evaluation prompts are the recaptions of the held-out sprites, so the evaluation prompts follow the training caption distribution.

C GUIDANCE BASELINES: EXACT SETTINGS

All baselines replace CFG in Eq. equation 1, on the same model, sampler, prompts and initial noise. Random draws for ICG and TSG come from a separate generator, so that switching methods does not change the sampler noise.

- **CDG** (Han et al., 2026), $R_{\text{deg}} = 1$: every position covered by the tokenizer’s attention mask (BOS, words, first EOS) takes the embedding of the empty caption at that position, while the remaining padding positions keep the caption’s states; the bucket is unchanged. On our recaptions, on average 16.3 of 77 positions are content, and no prompt fills all 77 (for such a prompt CDG would equal CFG).
- **ICG** (Sadat et al., 2025b): $\hat{c} = \text{std}(c) \cdot n$ with $n \sim \mathcal{N}(0, I)$ of the caption embedding’s shape, redrawn at every step; the target bucket is kept. **ICG-label**: the caption is kept and the bucket is drawn uniformly from $\mathcal{B}$ at every step.
- **TSG** (Sadat et al., 2025b): the time embedding (before the bucket embedding is added) is perturbed by $s(t/1000)^\alpha \text{std}(\cdot) n$ in all layers. “SD” setting: $s = 3$, $\alpha = 0.25$, applied for $t \geq 400$; “DiT” setting: $s = 2$, $\alpha = 1$, all t , $w = 2.5$.
- **SEG** (Hong, 2024): self-attention queries are Gaussian-blurred over their square layout (kernel size $\lceil 6\sigma \rceil + 1$, truncated to the map, reflect padding; $\sigma = \infty$ replaces the queries by their spatial mean), in the mid block (“mid”), in the down/up blocks at $R/2$ (“d1+u1”) or everywhere (“all”); $w = 3$. The “all, $\sigma = 1$ ” setting gives 122.52.
- **PAG** (Ahn et al., 2024): the self-attention map of the mid block is replaced by the identity.
- **Autoguidance** (Karras et al., 2024): the 10k snapshot under the target bucket and the same caption.

D EXTRAPOLATION-WEIGHT MEASUREMENTS

The “label” reference is the final model under the lower bucket, “snapshot” is the 10k snapshot under the target bucket (autoguidance), and “composed” is the snapshot under the lower bucket. All three

Table 8: MSE-optimal extrapolation weight w^* on forward-noised data (v7h; 10 timesteps from $t = 10$ to 990; 3,000 sprites; ranges over timesteps) and the relative denoising loss of the guided prediction at $w = 1.5$.

Resolution / rung	Data	label w^*	snapshot w^*	composed w^*	$L(1.5)/L(1)$
16 / 12	held-out	0.88–1.04	0.83–1.07	0.85–1.07	1.02–1.88
16 / 12	training	0.94–1.00	1.00–1.07	0.99–1.07	1.02–1.90
20 / 16	held-out	0.92–1.02	0.88–1.05	0.89–1.06	1.01–1.81
24 / 16	held-out	0.91–1.02	0.85–1.05	0.88–1.04	1.03–1.95

are evaluated on the same x_t and caption. On self-rolled states (2,000 held-out sprites), a sprite is noised to k steps before a target position in the 100-step schedule (positions 25, 45, 65, 85, 95), and the deployed sampler (bare, or label-guided with $w = 2$) runs k steps. w^* is then measured in x_0 -space against the known sprite. Over all $k \geq 5$ points, the maximum w^* is 0.895 / 1.069 / 0.989 (label / snapshot / composed) at 16 px, 0.825 / 0.997 / 0.978 at 20 px and 0.819 / 0.888 / 0.876 at 24 px. No point reaches 1.3.

E RESULTS ON THE BLIP-CAPTION MODEL V7H

Table 9: 16px results on v7h (BLIP captions, BLIP evaluation prompts) next to v7r. Three seeds where $\pm$ is given, otherwise seed 0.

Method	NFE	v7h (BLIP)	v7r (recaptioned)
CFG $w = 4$ (default)	2	21.52 $\pm$ 0.47	20.18
Best CFG ($w = 1.5$)	2	12.49 $\pm$ 0.72	11.28 $\pm$ 0.54
Autoguidance, 10k snapshot	2	8.73 $\pm$ 0.26	9.78 $\pm$ 0.50
Lower-bucket label reference	2	8.52 $\pm$ 0.29	8.86 $\pm$ 0.06
Composed reference	2	7.53 $\pm$ 0.19	8.50 $\pm$ 0.25
ICG, $w = 1.25 / 1.5$	2	8.88 / 11.29	7.68 / 7.46 $\pm$ 0.37
Internalized, $w = 1.5$	1	7.23 $\pm$ 0.34 (7.63 / 7.02 / 7.05)	7.23 $\pm$ 0.45
Internalized, $w = 1.25 / 2$	1	8.77 / 9.56	–
Internalized ($w = 1.5$) + CFG 1.25 / 1.5; ($w = 1.25$) + CFG 1.5	2	8.07 / 8.16 / 8.34	–
Higher-bucket reference, buckets 20 / 24 / 64, $w = 2$	2	17.14 / 17.27 / 19.34	–

On v7h the internalized model beats the composed rule at every seed (7.63 / 7.02 / 7.05 vs. 7.67 / 7.32 / 7.61). Higher buckets are harmful at every resolution: at 12 px, bucket 16 gives 14.45 vs. best CFG 9.71; at 20 px, buckets 24 / 32 give 46.62 / 59.13 vs. 39.53; at 24 px, bucket 32 gives 101.67 vs. 67.34 (seed 0).

What the lower-resolution belief looks like (v7h, 16px). Sampling each reference on its own (median unique colours over opaque pixels, fraction of exactly equal adjacent pixel pairs, mean adjacent $|\Delta RGB|$) gives 78 / .015 / 21.6 for the 12 px belief, against 73 / .030 / 32.0 for the strong model (CFG $w = 4$) and 34 / .202 / 30.2 for real sprites. The 24 px and 64 px beliefs have contrast 31.9 and 40.9. The 12 px belief is thus not a simpler image with fewer colours; it is a lower-contrast rendering of the same layout. A contrast reduction that is *not* structured like the belief does not work: shrinking the strong prediction’s own $\hat{x}_0$ contrast towards its mean and using that as the reference gives 74.8–444.4.

F COST

G SDXL SIZE-CONDITIONING PILOT

H CAPTIONS OF THE QUALITATIVE FIGURES

Held-out prompt indices 1, 4, 5, 9, 10, 31, 42, 44, 55, 58 (left to right in Figures 1 and 2):

1. Ear of corn with yellow kernels sitting in green leafy husks

Table 10: Wall-clock time for 1,000 samples at 16 px (A100-80GB, exclusive; 100 DDPM steps; batch 500; including model loading), measured on v7h. The internalized model has the same architecture plus 32,768 parameters and runs at 1 NFE; it was not timed separately.

Configuration	NFE	s / 1,000 samples	peak memory (MiB)
One pass per step (no guidance; internalized model)	1	42.5	4951
CFG	2	73.8	4951
Lower-bucket label reference	2	72.7	4951
Autoguidance / composed reference (snapshot in memory)	2	73.3 / 74.0	5507
SD- π XL (Binninger & Sorkine-Hornung, 2024), one 16 px sprite (shared node)	–	29,500 per sprite	–

Table 11: SDXL base 1.0 on the first 1,000 captions of COCO-30k shard 0 (1024 px, Euler 30 steps, identical initial noise). FD with DINOv2 ViT-L/14 CLS (224 px, bicubic) against real images 1,000–2,999; floor (real images 0–999) 507.05; CLIP ViT-L/14 100 cos (real 25.91). “orig” denotes the `original_size` condition of the weak branch.

Setting	FD	CLIP
CFG $w = 3 / 5 / 7$	662.22 / 644.18 / 648.44	26.15 / 26.59 / 26.82
CFG $5 + 1 \times (e_c - e_{c, \text{orig}=512})$	644.13	26.62
CFG $5 + 1 \times (e_c - e_{c, \text{orig}=256})$	658.98	26.46
CFG with <code>negative_original_size 512</code> (common practice)	650.98	26.46
Size reference only, orig 512, $w = 3$ (no CFG)	969.68	21.26

2. Magic wand with a brown wooden handle and a red orb with white highlights
3. Goblin, green skin and yellow eyes, holding a large brown wooden club
4. Green bipedal creature with a wide mouth and a visible pink tongue
5. Magnifying glass with a golden handle, white frame, and glowing green lens
6. Slime monster, vibrant magenta blob, two tiny dark green rectangular eyes
7. Glass potion bottle, brown cork, filled with bubbling dark purple liquid, thick outline
8. Cowboy wearing a wide-brimmed brown hat, brown vest, and a yellow star
9. Large fire, white-yellow core, intense orange glow, dark red jagged outlines
10. Baked pie, round golden-brown crust with a dark brown filling inside a dish

I PENDING RESULTS

[TODO: Remove this section before submission.] Items still running when this draft was written: the internalized model on v7r8 at 12 px (three seeds) and its 16 px regression check; the label reference on v7r8 at $w = 3/3.5$ and seeds 1–2; the third seed of the internalized model at 24 px; CLIP scores for the internalized + ICG configuration, plain GFT, the 20/24/12 px final configurations and the external systems; SDXL + Pixel Art XL LoRA (and its PixelOE variant) on the $n = 200$ protocol; SD- π XL sprites on the recaptioned prompts (qualitative only).